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Gait & Posture 23 (2006) 1–8
GAIT & POSTURE
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Relationship of spasticity to knee angular velocity and motion during gait in cerebral palsy

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Accepted 16 October 2004

Abstract

This study investigated the effects of spasticity in the hamstrings and quadriceps muscles on gait parameters including temporal spatial measures, knee position, excursion and angular velocity in 25 children with spastic diplegic cerebral palsy (CP) as compared to 17 age-matched peers. While subjects were instructed to relax, an isokinetic device alternately flexed and extended the left knee at one of the three constant velocities 30°/s, 60°/s and 120°/s, while surface electromyography (EMG) electrodes over the biceps femoris and the rectus femoris recorded muscle activity. Patients then participated in 3D gait analysis at a self-selected speed. Results showed that, those with CP who exhibited heightened stretch responses (spasticity) in both muscles, had significantly slower knee angular velocities during the swing phase of gait as compared to those with and without CP who did not exhibit stretch responses at the joint and the tested speeds. The measured amount (torque) of the resistance to passive flexion or extension was not related to gait parameters in subjects with CP; however, the rate of change in resistance torque per unit angle change (stiffness) at the fastest test speed of 120°/s showed weak to moderate relationships with knee angular velocity and motion during gait. For the subset of seven patients with CP who subsequently underwent a selective dorsal rhizotomy, knee angular extension and flexion velocity increased post-operatively, suggesting some degree of causality between spasticity and movement speed.
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Keywords: Stiffness; Kinematics; Angular velocity; Stretch responses; Diplegia

1. Introduction

Due to the multi-dimensional nature of the motor disorder in spastic cerebral palsy (CP), several factors interact to produce gait limitations in these patients. Motor coordination during walking and other motor tasks may be compromised by 'negative' features, such as restrictions in muscle length, reduced strength, poor selective control and by the 'positive' sign of spasticity. Patients with spastic CP, by definition, have increased velocity-dependent resistance to passive stretch, presumed to interfere with active motion. Spasticity may be focal (a single or only a few joints) or generalized (multiple joints). It has been proposed that children with CP are constrained in movement speed by spasticity, in that they 'avoid' or are prevented from achieving faster velocities so as not to elicit a stretch response [1]. While spasticity, when present, clearly interferes with passive muscle stretch in the relaxed state, the precise role of spasticity in disrupting voluntary or active motion remains a topic of considerable debate and clinical importance.

A stretch response is a normal protective mechanism that occurs when a resting muscle is lengthened rapidly or forcefully. A person is considered to have 'spasticity', when the threshold for this response is lowered and can be elicited much more readily, even on passive examination. The force and speed of a manual evaluation, such as performed when administering the Ashworth test for grading spasticity, typically underestimates those conditions which would

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doi:10.1016/j.gaitpost.2004.10.007

likely produce stretch responses in persons with intact neuromotor systems. On the other hand, instrumented systems, which can accelerate quickly and can safely move the joint at more rapid speeds are able to detect stretch responses in persons both with and without spasticity. The former group is often distinguishable by the slow speeds at which these responses have been shown to occur [2].

1.1. Relationship of spasticity to active movement

For decades, spasticity was considered to be the primary culprit in producing the motor deficit in cerebral palsy. While scientific evidence has failed to confirm this strong presumed link and suggests that spasticity has a more modest or perhaps even negligible effect on motor function [3], interventions aimed at alleviating spasticity such as selective dorsal rhizotomy, intra-thecal baclofen pump implantation and botulinum toxin injections, are still frequently prescribed. Conflicting reports on the role of spasticity during active movement have been noted in the literature for decades and no consensus or resolution has yet been reached.

Similar to the goals of this study, Norton et al. [4] specifically examined the relationship between spasticity at the knee and gait speed in 15 adults with hemiplegia, 10 in whom the deficiency was due to a cerebrovascular accident and five, who had sustained a head injury. Their measure of spasticity was the area under the smoothed and rectified electromyography (EMG) curve of the antagonist muscle during passive knee flexion and extension in a sinusoidal (changing velocity) and ramp (constant velocity) condition and they correlated these values with the fastest voluntary walking speed of the patients. None of the correlation values reached significance, although the authors felt that a trend was evident between flexor muscle spasticity and gait speed that might have reached significance with a greater number of patients with more homogeneous disorders. Sahrmann and Norton [3] in a later study relating spasticity in the elbow musculature to rapid alternating movements find more conclusively that spasticity was related to movement speed in the upper extremity. In this study, a relationship was only found between elbow flexor, but not extensor, spasticity and the speed of movement. Ada and co-workers [5] compared the magnitude of stretch responses at the ankle during passive (patient relaxed) and active (patient attempting movement) conditions in patients with strokes compared to neurologically normal controls. They found that two-thirds of the patient group had heightened stretch responses during passive lengthening with none of the controls showing stretch responses. The patient group did not show higher levels of antagonistic resistance than controls during the active condition, leading these authors to conclude that spasticity has little to do with impairing muscle activation during gait.

Crenna [6] proposed that the potential effect of spasticity on active motion was dependent on the type of movement being performed, irrespective of its velocity. In studies on CP gait, he noted that abnormal stretch responses were more easily elicited during lengthening contractions around the time of ground contact than at other points in the cycle, and he hypothesized that this may occur because floor contact is a critical event that may be more dependent on supraspinal rather than peripheral control. He further suggested that disruptions in movement in other parts of the cycle may be attributable to weakness, co-contraction or passive muscle properties, rather than to heightened stretch responses.

Knutsson and co-workers [7] agreed spasticity could impair voluntary movement if the spastic muscle was lengthening. They showed clear differences in the amount and velocity-dependence of restraint from a spastic antagonist during concentric action of the agonist versus eccentric action of that same agonist, when the spastic antagonist was shortening. They also pointed out that the ability of the agonist to produce force, independent of antagonist restraint, was concurrently disrupted in patients with spasticity, and that both impairments worsened with increasing the velocity. Similar findings of relatively greater concentric compared to eccentric weakness and greater antagonist resistance at faster test velocities have been reported in CP [8]. Sinkjaer [9] proposed that movement is disrupted in patients with ankle spasticity in part, because they lack modulation of reflexes during gait as seen in healthy controls. He further noted that passive stiffness is increased in these same patients, also contributing to the motor deficit.

While some of these findings seem to indicate that treatment of spasticity is unwarranted or would be only marginally effective if the goal of the intervention is to improve movement, others suggest the possibility that therapeutic intervention aimed at reducing spasticity and/or an increasing muscle strength may improve the ability to perform movements, particularly at faster speeds. However, many of these specific causal relationships have not been verified.

1.2. Gait patterns in 'spastic' cerebral palsy

Patients with cerebral palsy have reduced gait velocity at free and fast speeds that is in proportion to their level of neurological involvement [10,11]. Stride lengths are restricted and cadence is often increased. Angular velocity at the hip, knee and ankle, normalized by stride time to remove the effects of cadence, has been shown to be reduced in a group of patients with CP compared to normal [12]. Kinematic parameters also differ in these patients and for reasons beyond diminished velocity alone. Specifically at the knee joint, two kinematic patterns have been identified in this population, which have been presumed to be directly related to spastic responses in the hamstrings and quadriceps muscle groups. The first is 'crouch' gait which is characterized by excessive knee flexion at initial contact and during mid-stance when the knee typically reaches its

maximum extension during the gait cycle. Inability to fully extend the knee disrupts the normal plantar-flexion knee extension couple and necessitates increased muscular effort to sustain this posture. Muscle tightness, compromised ambulatory ability with age and growth and long term joint deterioration may also ensue if this pattern persists. A second prevalent pattern is 'stiff-knee' gait in which total knee excursion is limited, particularly peak knee flexion during swing. In contrast to crouch, the functional implications of this pattern are primarily seen in swing with a disruption of foot clearance that can affect the smoothness and stability of the gait pattern. The etiologies of these knee patterns remain incompletely understood despite multiple hypotheses. Strengthening programs targeting these and other muscles can lead to significant improvements in temporal spatial and kinematic parameters in these patients, inferring some degree of causality [13]. Restricted muscle length at the knee and adjacent joints has also been shown to produce or exacerbate these and other kinematic abnormalities [14].

1.3. Hypotheses

The primary purpose of this investigation was to examine the relationship of heightened stretch responses (spasticity) in the quadriceps and hamstrings muscles as detected during passive motion, to gait parameters including knee angular velocity, which reaches its maximum values during swing, and knee position and excursion throughout the gait cycle. Using an instrumented measure in combination with electromyography that can more precisely identify the presence and magnitude of stretch responses, we sub-grouped the children with CP and hypothesized that the group of children with stretch responses at relatively slow velocities would show differences in gait from children with CP who did not exhibit stretch responses at these velocities. All children with CP were expected to show differences from age-matched peers, but these were projected to be more pronounced in the 'spastic' sub-groups. More specifically we proposed that (1) peak knee extension velocity would be slowest in swing for those with hamstrings stretch responses and peak knee flexion velocity in swing would be least in those with quadriceps stretch responses, (2) hamstring spasticity would specifically limit stride length and knee extension, and (3) quadriceps spasticity would specifically limit total excursion and maximum flexion achieved in gait.

Causal relationships were explored in a limited subset of these patients who subsequently underwent selective dorsal rhizotomy to provide greater insights on the specific effects of reducing spasticity on gait. Selective dorsal rhizotomy has been shown to predictably reduce spasticity without a measurable effect on muscle strength [15] or isolated motor control [16] in the short term. Therefore, this intervention provides an opportunity to evaluate the causal effects of spasticity on movement speed with the hypothesis that knee joint angular velocity would increase after SDR.

2. Methodology

2.1. Subjects

In all, 42 children, 25 ambulatory children with spastic diplegic CP and 17 age-matched children without CP gave their written assent or consent along with that of their legal guardian to participate in this protocol. The mean ages were 10.9 ± 4.3 years and 11.2 ± 2.5 years in the respective groups. In the group with cerebral palsy, seven subsequently underwent a selective dorsal rhizotomy, and also participated in a second identical assessment post-operatively, approximately 8 months after surgery.

2.2. Procedures

All participated in the following assessments performed or recorded on the left side only with the exception of gait velocity: (1) physical examination of passive muscle length and flexibility at the knee joint, knee flexor and extensor Ashworth Scores and Ely test for rectus femoris spasticity, (2) measurement of resistance torque during passive knee flexion and extension on an isokinetic device at three velocities 30°/s, 60°/s and 120°/s, while the patient was instructed to remain completely relaxed, (3) 3D gait analyses of temporal-spatial and kinematic parameters of free-speed gait, (5) electromyographic assessment of the hamstrings and quadriceps muscle groups during all isokinetic and gait trials with pre-gelled surface electrodes (Medicotest A/S, Denmark) placed over the muscle bellies of the biceps femoris and rectus femoris muscles at an inter-electrode distance of approximately 1–2 cm, depending on the size of the child. The rectus femoris differs from the vasti in its action as a two-joint muscle, and can be distinguished even further in CP by the prevalence of spasticity [17].

2.3. Data analysis

From the passive isokinetic trials, two values were obtained for each of the two muscle groups at each of the three velocities: peak resistance torque and the slope of the constant velocity portion of the resistance torque by angle curve, as a measure of muscle stiffness. Kinematic parameters evaluated were total knee excursion, and the maximal and minimal knee position during the gait cycle. Peak knee angular velocity, which invariably occurred during swing, was computed from the kinematic data, but was not normalized by stride time as described previously [11] to allow for direct comparison to isokinetic velocities.

The sample with CP was sub-grouped based on the presence or absence of hamstrings and/or quadriceps spasticity. Angular velocities were compared across groups (CP spastic, CP non-spastic and age-matched group without CP) using oneway multiple analysis of variance (MANOVA) procedures with $p < 0.05$. In the surgical subgroup, pre-operative versus post-operative comparisons were per-

formed using repeated measures MANOVA procedures. Pearson correlations were used to examine the relationships among passive isokinetic values, knee position and excursion, knee angular velocities and gait temporal spatial parameters were only in the subjects with cerebral palsy. Partial correlations removing the co-linearity of velocity of forward progression with knee angular velocity were also performed. Surgical changes in angular velocity with changes in other measured gait parameters were also correlated in that subset.

3. Results

3.1. Stretch responses

Of the 25 children with CP, 18 demonstrated hamstring stretch responses, 10 demonstrated quadriceps responses, 9 showed responses in both muscles, and 6 in neither. Specifically, EMG activation was noted in those muscles each time they were passively stretched at at least one of the three velocities tested. The six patients with CP without any stretch responses at isokinetic speeds tested comprised the 'non-spastic' CP group for all comparisons. Although this delineation seems to be in contradiction to the diagnosis of spastic CP, this merely means that stretch responses were not detectable at the knee joint tested at speeds up to 120°/s in those patients. This does not mean that stretch responses could not be detected at speeds greater than this or at other joints that would be indicative of the presence of spasticity.

Because of the possible confounding effect of having spasticity in both muscles, the hamstring group results were also analyzed as a hamstring only ($n = 9$) and hamstring + -quadriceps ($n = 9$) spastic group. Since the quadriceps group was comprised of those who had spasticity in both muscles in 9 of 10 cases, it was not sub-divided further. No child in the comparison group without CP showed stretch responses in either muscle at any test velocity. Stretch responses, when found, were detected at 30°/s in all, but there were two patients who instead demonstrated them first at 60°/s. All in the spastic groups had responses at all test speeds higher than that of the initial response.

3.2. Gait

Peak angular velocity in flexion and extension differed significantly in both CP groups compared to the non-CP controls (see Fig. 1a). Those with quadriceps spasticity had a significantly slower peak flexion velocity than the CP group without spastic responses. Peak extension velocity in the entire group of those with hamstring spasticity (includes those with and without concurrent quadriceps spasticity) was not significantly slower than in those with CP who did not demonstrate spasticity; however, the sub-group with hamstrings + quadriceps spasticity was significantly slower than the non-spastic CP group (see Fig. 1b).

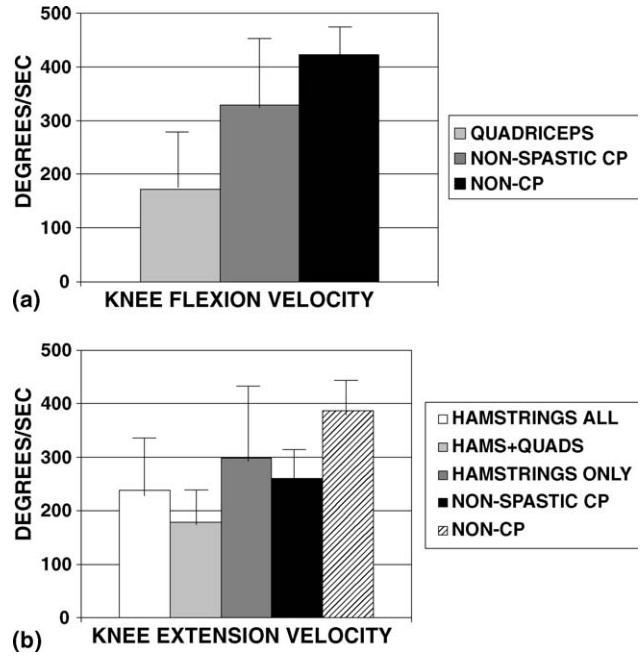

Fig. 1. (a) Mean peak knee flexion velocity with standard deviation bars plotted for the group with quadriceps spasticity, the CP group with no hamstring or quadriceps spasticity, and the non-CP comparison group. (b) Mean peak knee extension velocity with standard deviation bars plotted for the group of all who had hamstrings spasticity ($n = 18$), the subgroups of those who had hamstrings + quadriceps spasticity ($n = 9$) or hamstrings spasticity only ($n = 9$), and the non-CP comparison group.

For the entire group with hamstring spasticity compared to the CP group without spasticity, knee minimum was higher (29.0° compared to 17.0°), and knee excursion (34.0° compared to 45.9°), gait velocity (0.67 m/s compared to 1.03 m/s), and stride length (0.73 m compared to 1.01 m) were less. All of these parameters in the spastic group also differed significantly from the non-CP comparison group; however, the CP group without spasticity differed significantly from the comparison group only in total knee excursion (45.9° compared to 54.8°).

Comparing the quadriceps spasticity group to the non-spastic group with CP, velocity of forward progression was slower (0.56 m/s compared to 1.03 m/s), stride length was shorter (0.63 m compared to 1.01 m), and knee excursion was less (33.0° compared to 45.9°); position did not differ within CP subgroups. Knee excursion was significantly less and knee minimum was higher (more crouched) in the CP spastic group compared to the non-CP comparison group. As depicted in Fig. 2a and b, it is these differences in maximum knee extension rather than flexion, and the resultant effects on excursion, which best distinguishes the groups.

The results of using the Ely test to determine quadriceps spasticity produced statistically similar results as using isokinetic tests to differentiate spastic versus non-spastic groups. The Ely test results were identical to the isokinetic/EMG determination in two/three of the cases; however, in the remaining third, the clinical test tended to identify more

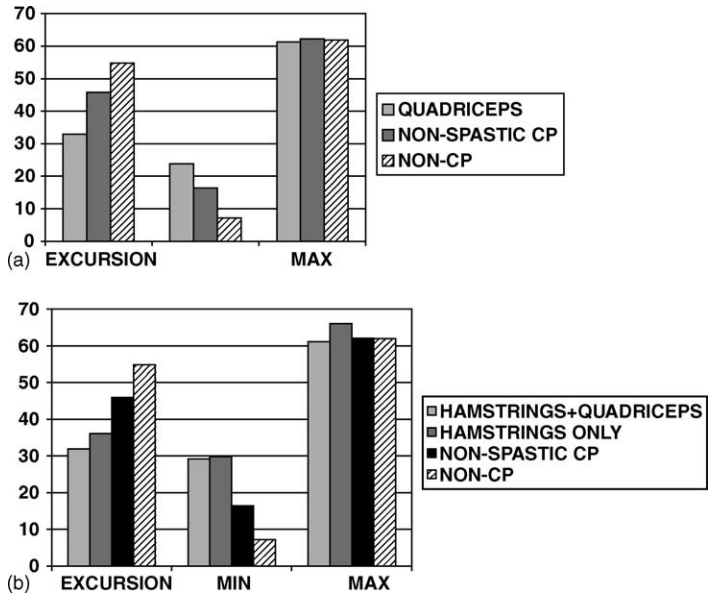

Fig. 2. Mean total knee excursion and positions (min = minimum flexion, max = maximum flexion) during the gait cycle when walking at a self-selected speed, plotted by groups based on the presence or absence of spasticity in the (a) hamstrings and/or (b) quadriceps muscles. The groups differ primarily in knee minimum or degree of crouch, reflected by changes in excursion, rather than knee maximum which is the maximum flexion achieved during swing.

patients as having quadriceps spasticity than instrumented/EMG measures.

3.3. Surgical subgroup

In the seven children who subsequently underwent selective dorsal rhizotomy, four showed stretch responses pre-operatively in the quadriceps and hamstrings at 30°/s, two others had these only in the hamstrings also at 30°/s and one did not show stretch responses in either muscle group up to 120°/s. No stretch responses were detectable post-operatively in any of the subjects at any test speeds. Both peak knee flexion and extension angular velocity at the freely chosen gait speed increased markedly and significantly post-operatively, with a mean increase of 83.5°/s for flexion and 52.6°/s for extension. Temporal spatial parameters of velocity and stride length showed no significant mean change in this sample. Similarly, knee excursion and position during gait did not change significantly, but did show a tendency towards improvement in total excursion that was achieved by both greater extension in stance and greater flexion in swing (+5.3°, +2.6°, and +2.7°, respectively).

3.4. Relationships of spasticity and gait parameters in cerebral palsy

Maximum extension velocity was correlated with hamstring and quadriceps stiffness at 120°/s ($r = -0.47$ and -0.41, respectively). Maximum flexion velocity was correlated with quadriceps and hamstring stiffness at 120°/s ($r = -0.50$ and -0.39, respectively). This indicates that increasing stiffness in either/both the agonist and antagonist is negatively related to angular velocity. The only significant relationship of instrumented measures of spasticity to knee motion was stiffness in quads at 120°/s with knee excursion ($r = -0.39$). Mean stiffness values were typically lower in the non-spastic CP group compared to the spastic groups, but the differences were not significant. All CP groups, even the one without stretch responses, had significantly higher stiffness than the control group (see Fig. 3).

In the group that underwent selective dorsal rhizotomy, pre-operative to post-operative changes in knee flexion velocity were directly and significantly correlated with changes in maximum knee flexion achieved during gait ($r = 0.89$, $p < 0.01$).

4. Discussion

Identifying stretch responses during passive motion in a relaxed subject is straightforward; however, trying to identify these responses during voluntary motion is much more problematic unless the muscle being lengthened is normally silent or the intended motion is actually halted or reversed by an involuntary response in the antagonist. An example of an abnormally heightened stretch response can be seen at the ankle in many children with CP who contact the ground on their forefoot and then experience an early heel rise rather than increased dorsiflexion which normally occurs. The speed and direction of movement as well as the state of voluntary muscle activation in the agonist and the antagonist may influence the degree to which spasticity may impede voluntary movement. If both muscle groups crossing a joint are relaxed during a specific motor task, it seems reasonable to assume that the response may be similar to that

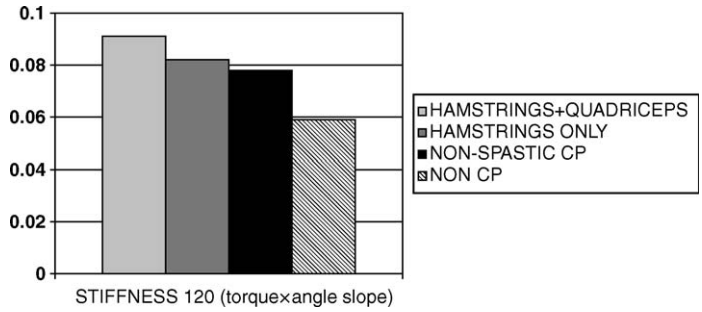

Fig. 3. Mean stiffness in the hamstrings muscles, calculated as the slope of the torque × angle curve, at the isokinetic velocity of 120°/s plotted for the subgoups with hamstring + quadriceps spasticity, hamstring spasticity only, the CP group with no hamstring or quadriceps spasticity, and the non-CP comparison group.

which is seen when muscles are stretched passively. If the agonist is active during lengthening of the antagonist, reciprocal inhibition may serve to limit antagonist activity, although this assumption may not be valid in some individuals with a brain injury [9]. Voluntary antagonistic activity during its lengthening (e.g. co-contraction) or passive muscle properties may increase the total resistance, making the contribution of spasticity to the EMG indistinguishable. In normal swing, knee flexor and extensor muscles are active in both the flexion and extension phases, which obscures EMG evidence of stretch responses, making inference necessary. In this study, we attempted to evaluate the influence of stretch responses at the knee during gait by predicting their consequences such as reduced active motion and angular velocity at that joint.

To summarize results, in this sample, the sub-groups of children with cerebral palsy were differentiated from those without CP by slower peak knee angular velocities and less total excursion during gait. Quadriceps spasticity, or the identification of stretch responses at velocities less than 120°/s in the quadriceps (9/10 who also had these in the hamstrings), differentiated the CP groups in knee flexion angular velocity and total excursion, as hypothesized, and also in gait velocity and stride length, with all values being lower in the spastic sub-group. Contrary to the hypotheses, maximum knee flexion was not less in the spastic versus non-spastic group. Hamstring spasticity, or the identification of stretch responses in the hamstrings at velocities less than 120°/s (with or without quadriceps responses), differentiated the CP groups in maximum extension (greater crouch), stride length, gait velocity and excursion with all was more limited in the spastic group. Interestingly peak knee extension velocity was not significantly slower than the non-spastic CP group in those with hamstring spasticity only or when these subjects were combined with the hamstrings + quadriceps spasticity group. Peak angular velocities increased as a result of selective dorsal rhizotomy without concurrent significant changes in gait motion or temporal spatial parameters.

The occurrence of stretch responses did not halt passive isokinetic knee motion in CP, since motion continued in all cases during this motion after the stretch response was elicited. Stretch responses also did not appear to arrest knee motion during gait in these patients with CP when they reached or surpassed the threshold velocity of the passive test response, since the peak angular velocity values in the spastic groups greatly exceeded the velocity at which these were detected, with mean peak angular velocity values in those with stretch responses in the respective antagonistic muscles being 172.4°/s for knee flexion velocity and 239.0°/s for knee extension velocity.

Slowing of knee motion was correlated with increasing stiffness values in both the agonist and antagonist. Spasticity has been shown previously to increase stiffness in cerebral palsy, with the highest values and greatest differences from non-spastic groups seen at the highest speeds of muscle lengthening. Therefore, the correlations of stiffness at only at the highest test speed 120°/s, with angular velocity provides some support for the possibility that increased stiffness caused by spasticity may limit movement speed. The potential effects of stiffness of non-reflex origin on angular velocity cannot be dismissed and warrant further study, since even those with CP but without evidence of heightened stretch responses in the knee musculature tend to have greater stiffness values, slower angular velocities and less motion than age-matched controls.

Several factors may have skewed results and limited interpretations in this investigation. The first is the co-presence of heightened stretch responses across the two sides of the joint in many of the patients. Spasticity in the hamstrings, regardless of whether or not the quadriceps were involved as well, did seem to differentiate groups in severity of mid-stance crouch. Unfortunately, the unique role of quadriceps spasticity alone in stiff knee gait could not be determined here due to the confounding factor of hamstring spasticity in those patients. While the presence of stretch responses in both agonist and antagonist muscle pairs confounds the issue, it is important clinically to note that the combination seemed to pose even greater limitations on gait.

Neither the stiff knee gait pattern nor the crouched gait pattern improved significantly in the surgical sub-group, and while the power was low due to the sample size, the effective size produced here was also very small and unlikely to be of clinical significance if the same trend were to be upheld in a larger study. This stimulates re-evaluation of the magnitude of the contribution of spasticity or co-spasticity, to stiff or

crouched knee patterns. Changes over time leading to restrictions in muscle length, which may be mediated in part by spasticity, or reductions in muscle strength may alternatively be responsible for the persistence of these patterns even after surgical reduction in spasticity.

The second confounder is that the rectus femoris and the long head of the biceps femoris are two joint muscles; therefore, each can exert forces to either produce or restrain movement at the hip as well as the knee, often in conflicting ways. For example, it is difficult to explain why hamstring spasticity alone led to nearly equivalent reductions in excursion as quadriceps + hamstrings spasticity in this sample. One hypothesis that warrants further exploration is the possibility that hamstring spasticity may reduce knee excursion by restraining hip flexion through its other role as a hip extensor and thereby, limit knee flexion secondarily even in the absence of quadriceps spasticity. Similarly, rectus spasticity may restrain or slow hip extension during gait in addition to knee flexion.

A further point that makes interpretation difficult is that, although subjects were grouped by the presence or absence of stretch responses at moderately fast velocities, the 'spastic' versus 'non-spastic' distinction made here may only have isolated a more neurologically impaired subset. It has been shown previously that strength and motor function tend to be affected to a greater degree in more spastic patients, and these factors could have contributed to or produced the greater gait deficits in this group [18].

While the results from the analysis of the surgical subgroup are very preliminary, the change in angular velocity was very pronounced and occurred in the absence of significant change in self-selected velocity of forward progression. Interestingly, an earlier study failed to detect a change in knee angular velocity as a result of muscle-tendon lengthening despite a significant increase in knee extension, stride length and forward velocity [12], suggesting that surgical increases in muscle length may affect active joint excursion but not speed. It is also possible that the changes seen here in angular velocity may not have been due to the reduction of spasticity, and may be attributable instead to other aspects of the post-operative course such as intensive strengthening programs typically prescribed after SDR, since strengthening alone has been shown to increase cadence, gait velocity and hip and knee motion during gait [11,13,19]. Further research is needed to evaluate the unanswered questions such as the effects of quadriceps spasticity alone, the consequences of spastic responses at the hip or ankle joints, and and the effects of other spasticity interventions or of strengthening protocols such as isokinetic training or electrical stimulation on the ability to increase joint angular velocity.

5. Conclusions

Those who show heightened stretch responses in both the hamstrings and quadriceps have reduced angular velocities compared to those with CP, who did not demonstrate stretch responses in this protocol. A similarly definitive statement cannot be made about those who have stretch responses in only a single muscle based on the results shown here. Previous work had shown that patients with CP who do not demonstrate heightened stretch responses may still have increased passive muscle stiffness and weakness that can also contribute negatively to muscle performance. Our results here further suggest that spasticity reduction may have a direct causal link to movement speed in CP, with a more limited or negligible effect on kinematic patterns at the knee.

Acknowledgements

This work was funded by the National Center for Medical Rehabilitation Research at the NIH and by the United Cerebral Palsy Research and Education Foundation.

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